

Exercise 2 User and Group Concepts in Linux/Unix

Learning objectives:

- Find out the users, authorized users, and groups of a Linux system using relevant commands
- Create and delete users, authorized users, and groups in a Linux system using shell commands
- Explain permissions for users and groups of files and directories
- Identify configuration files for users and groups
- Describe and explain the structure of configuration files

Legend:

 Reading material to be studied before the exercise

 Questions/tasks to be done before the exercise

 Tasks to be done during the exercise

2.1 Preparations for the Exercise

The preparations and questions are to be completed before the exercise session. The exception is this first exercise, as you have not yet had a chance to do any reading. They are intended to help you with the tasks you will work on during the exercise session itself.

 To prepare, read

- https://en.wikibooks.org/wiki/A_Quick_Introduction_to_Unix/Permissions
- <https://help.rc.unc.edu/how-to-use-unix-and-linux-file-permissions/>
- <https://www.linode.com/docs/guides/linux-users-and-groups/>
- https://www.christianbaun.de/BTS1314/Skript/folien_bts_uebung_02_WS1314.pdf (in German)

Alternatively, you can watch the following videos for preparation

- Linux Crash Course - Managing Groups

- Linux Crash Course - Managing Users
- Linux Crash Course - Understanding File Directory Permissions

For more theoretical background, the following sources are recommended:

- Chapter 7 of Ahmed F. Sheikh „CompTIA Linux+ Certification Companion”
- Chapter 2 Basics in the book „The Unix Operating System” (in German)
- Chapter 6 „Linux-UNIX Programming” (in German)



Introductory Questions:

Which of the following commands do you use to change the password of the current user in the shell?

- ☐ useradd
- ☒ passwd
- ☐ chpwd
- ☐ pwd

What does the command `su` stand for?

- ☐ See Unix
- ☐ See You
- ☒ Substitute User
- ☐ SUDoku

Which command can you use to display the users currently logged in on the Linux/Unix system?

Research which configuration files list the authorized users and groups on the system! And how the files are structured.

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Explain and note how you can find out in a UNIX shell:

- a) Which users are currently logged in on the system.

- b) Which groups you belong to as the current user.

- c) As which user you are currently logged in.



Important to Start:

Keep in mind that everyone can and should log in to the provided system. Please make sure to copy all personal data to your personal storage areas, to which only you have access, before the end of the exercise.

If you change or add users and passwords, please **make sure to keep the default user pi** with its original password, usually empty. Otherwise, you make the environment unusable for others, and we will have to reinstall the Raspberry Pi systems afterward.

2.2 Users and Groups

❖ Find out which users and which user groups are created on the Linux system. Copy (if necessary, in excerpts) the content of the files into the following field and explain the individual sections.

Users:

User groups:

2.3 Create Users and Add Groups, Add Users to Groups

- a) Create a new user **opsysuser** with the home directory **/home/opsysuser!** What do you need to do beforehand?
- b) Create a new group **opsysexcercise** and add the user created above to this group.
- c) How do you verify that the user and group creation was successful? Perform this verification.

- d) With which UID was the new user created, and with which GID was the new group created?

- e) Log out and log back in as the newly created user.
- f) Switch to the home directory of the student user and try to create a file **test.txt**. What do you observe?

- g) Log out again and log back in as the student user.
- h) Delete the group **opsysexercise**, the user **opsysuser**, and the home directory **/home/opsysuser** from your system.